

Midterm - 10/29/2015

- Points pre-assigned to review questions and problems are just an indication of the final score: partial and unsatisfactory answer/solution will receive a fraction of the points, whereas excellent answers may exceed the indicated points.
- Use of books/notes is forbidden.
- Indicate name and UC ID number in ALL the solution sheets. Number the solution sheets and indicate total number of sheets (*e.g.*, first sheet 1/3, second sheet 2/3, third sheet 3/3).

R1 (10Points) What is the main source of packet loss in the (wired) internet?

R2 (15Points) List and describe the sources of delay in the Internet.

R3 (15Points) What is the main functionality of the network layer in the internet?

R4 (20Points) Why is circuit switching used instead of packet switching in the public telephone network? Why is packet switching more efficient in the Internet infrastructure?

R5 (20Points) Briefly describe the performance requirements of real-time multimedia applications. Why is Forward Error Correction preferable to packet retransmission?

R6 (20Points) Describe the leaky bucket mechanism. Why is this type of mechanism needed in priority queueing?